Skills Summary

Center, Spread, and Shape

Use this reference while completing your worksheet or when studying.

Skill 1 — Center: mean or median?

Mean = total ÷ count. It is a balance point, so every observation pulls on it by its distance from the centre — one extreme value moves it.

Median = the middle of the ordered list (average the middle two when the count is even). It is a counting point, so it does not care how far away the extremes are.

Rule: symmetric data ⇒ report the mean. Skewed data or an outlier ⇒ report the median, which is *resistant*.

Example: Report the centre of 3, 5, 6, 8, 9, 11, 40.
Step 1. One value sits far above the rest, so the data is not symmetric — that makes the median the summary to report, not the mean.
Step 2. Seven values, so the middle is the 4th: $3, 5, 6, \mathbf{8}, 9, 11, 40$. (The mean would be 11.71, higher than six of the seven values.)
⇒ median = **8**

Try it: Report the centre of 12, 14, 15, 17, 20, 22, 60.

check: median = **17**

Skill 2 — Spread: range or IQR?

Range = max − min. Only two observations, both extreme, so a single outlier can double it.

IQR = $Q_3 - Q_1$, the width of the middle half. Split the ordered data at the median; Q_1 is the median of the lower half and Q_3 the median of the upper half.

Rule: the IQR travels with the median and is resistant; the range and the standard deviation travel with the mean and are not.

Example: Find the IQR of 4, 6, 7, 9, 10, 12, 13, 30.
Step 1. Spread beside a resistant centre means the IQR, so split the eight ordered values into two halves of four and take the median of each half.
Step 2. Lower half $4, 6, 7, 9$ gives $Q_1 = 6.5$; upper half $10, 12, 13, 30$ gives $Q_3 = 12.5$; so $Q_3 - Q_1 = 12.5 - 6.5$.
⇒ IQR = **6** (the range is 26 — more than four times as big)

Try it: Find the IQR of 10, 12, 14, 18, 20, 24, 26, 70.

check: IQR = **12**

Skill 3 — Shape: read it off the two centres

Shape is **symmetric**, **skewed right** (long tail toward the high values), or **skewed left** (long tail toward the low values).

The tail pulls the mean and leaves the median:

- mean $>$ median $\Rightarrow$ skewed right
- mean $<$ median $\Rightarrow$ skewed left
- mean $=$ median $\Rightarrow$ symmetric

On a histogram, name the side the *tail* is on, not the side the peak is on.

Example: A data set has mean 14 and median 9. Name the shape.

Step 1. The two centres respond differently to a tail, so compare them rather than trying to picture the data.

Step 2. The mean sits above the median, and only far-off high values can lift a balance point away from a counting point.

$\Rightarrow$ $\mathbf{14 > 9}$, so the distribution is **skewed right**

Try it: A data set has mean 31 and median 40. Name the shape.

check: $\mathbf{31 < 40}$, skewed left

(!) Watch out: shape is decided *first*. Centre and spread are a matched pair chosen from it — mean with standard deviation, or median with IQR. Reporting a mean alongside an IQR, or quoting a mean for visibly skewed data, is the error that costs marks even when every calculation is right.